

CSC435 Introduction to the Linux Ecosystem

The Shell, SSH, Git, Virtual Machines

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Outline

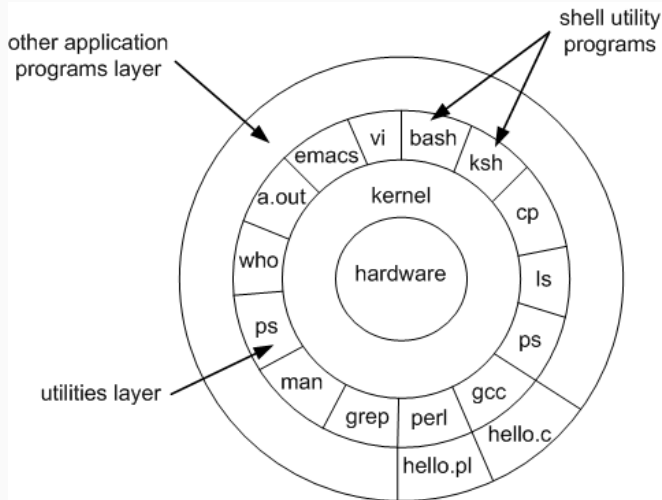
1. Introduction
2. The Shell
3. Secure Shell
4. Git
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Introduction

Operating System (Overview)

- acts as an intermediary between a user and the hardware;
- manages computer hardware and software resources;
- large and complex software, implemented in layers;
- variety in purpose, design and implementation;
- Processes/Compute, Memory, Input-Output;
- What does it mean for any of these resources to be "*intensive*"?

Operating System (Overview)



http://homepages.uc.edu/~thomam/Intro_Unix_Text/OS_Organization.html

The Shell

The Shell

- collection of utilities, text-based interface;
- allows users to communicate with the computer;
- allows programs to be started and tasks to be run;
- shell scripting;

- **sh** (Bourne shell) - basic shell, all UNIX systems;
- **ksh** (Korn shell) - complete high-level programming language;
- **bash** (Bourne Again Shell) - combines Korn Shell and C Shell (csh), default in most Linux distributions;
- **dash** (Debian Almquist shell) - fast, secure and minimalist shell;
- **zsh** (Z-Shell) - highly interactive and improved tab completion;

Unix/Linux commands

- file commands (ls, cd, pwd, rm, mv, cat, head);
- process management (ps, top, kill, bg, fg, jobs);
- file permissions (chmod, chown);
- system information (whoami, uname, man, du, df, free, which);
- compression (tar, gzip);
- network (ping, dig, wget, ssh, ifconfig, ip);
- build and install (make, cmake, automake, dpkg, rpm, gcc);
- shortcuts (Ctrl+C, Ctrl+Z);

Ubuntu Server Docs [*https://ubuntu.com/server/docs*](https://ubuntu.com/server/docs)

Intro to Linux

[*https://tldp.org/LDP/intro-linux/html/intro-linux.html*](https://tldp.org/LDP/intro-linux/html/intro-linux.html)

Command Line Interface

[*https://tldp.org/LDP/GNU-Linux-Tools-Summary/html/GNU-Linux-Tools-Summary.html*](https://tldp.org/LDP/GNU-Linux-Tools-Summary/html/GNU-Linux-Tools-Summary.html)

Getting the Shell

- Windows 10/11 ⇒ **Windows Subsystem for Linux (WSL2)**;
- Mac, BSD, Solaris and Linux ⇒ **Terminal**;

How to Install WSL2

<https://learn.microsoft.com/en-us/windows/wsl/install>

Secure Shell

- cryptographic network protocol;
- secure channel over unsecured network;
- client-server architecture (SSH client and SSH server);
- OpenSSH is the most popular implementation;

Uses:

- login remote host;
- execute command on remote host;
- automatic login (passwordless login) to remote server;
- secure file transfer;
- forwarding and tunneling ports;
- forwarding X from a remote host;

Public-Private Key Setup

Linux Distro, MacOS, BSD, Solaris, Windows 10/11 and WSL2;

```
localhost$ ssh-keygen
```

```
localhost$ ls -l ~/.ssh
```

```
-rw----- id_rsa
```

```
-rw-r--r-- id_rsa.pub
```

```
localhost$ ssh-copy-id <username>@<remote-host>
```

```
localhost$ ssh -i ~/.ssh/id_rsa <username>@<remote-host>
```

```
localhost$ scp ~/.ssh/id_rsa.pub <username>@<remote-host>:id_rsa.pub
```

```
<remote-host>$ touch ~/.ssh/authorized_keys
```

```
<remote-host>$ chmod 644 ~/.ssh/authorized_keys
```

```
<remote-host>$ cat id_rsa.pub > ~/.ssh/authorized_keys
```

Git

- version-control system for tracking changes in files;
- fast, ensures data integrity;
- distributed, non-linear workflows;
- free and open-source;

Uses:

- source code management;
- coordinate work between multiple developers;
- GitHub, GitLab, Atlassian Bitbucket;
- the development of the Linux kernel;

Git Tutorial [*https://git-scm.com/docs/gittutorial*](https://git-scm.com/docs/gittutorial)

Git Book [*https://git-scm.com/book/en/v2*](https://git-scm.com/book/en/v2)

GitHub Tutorial

[*https://docs.github.com/en/get-started/quickstart*](https://docs.github.com/en/get-started/quickstart)

Clone the repository on your computer(s):

```
$ git clone <github git url>
```

Add and update files, commit changes and push modifications

```
$ git status
```

```
$ git add <file>
```

```
$ git commit -m "<message>"
```

```
$ git push origin
```

Get updates and pull modifications

```
$ git fetch origin
```

```
$ git pull origin
```

Virtual Machines

Virtual Machines Technologies

- effective resource sharing, higher flexibility and an increased degree of security;
- operating systems, programming languages, compilers and computer architecture;
- Process VMs: Multiprogramming, Emulators and Dynamic Binary Translators (Windows NT), Binary Optimizers, High-Level Language VMs (JVM and .Net);
- System VMs: Virtual Machine Monitors, Emulators, Co-designed Virtual Machines;

System Virtual Machines

Type-1, native or bare-metal hypervisors:

- Xen;
- Oracle VM Server;
- VMware ESX/ESXi;

Type-2 or hosted hypervisors:

- Linux QEMU and KVM;
- BSD bhyve;
- **Oracle VirtualBox;**
- **VMware Workstation;**

Light-Weight Virtualization:

- **Linux LXC/LXD;**
- BSD Jails;
- Docker;

Last remarks

Where there is a shell, there is a way.